The Two-Player $3n \pm 1$ Game: A Binary Normal Form and a Well-Founded No-Draw Proof

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`https://github.com/Grisha-Pochuev/3n-plus-minus-1-game`

10 August 2026

Abstract

Consider the impartial two-player game on positive odd integers in which a move from $n > 1$ chooses $3n-1$ or $3n+1$ and removes all powers of two; reaching 1 wins immediately. Arbitrary play need not terminate: for example $5 \to 7 \to 5$ is a legal cycle. The question is whether optimal play nevertheless has finite remoteness from every starting position. We conjugate the game to a binary normal form with an expanding move A and a strictly contracting move B, introduce constant-tail coordinates and a three-free source map, and rank marked proof configurations by retained source, finite proof-height tokens, and a typed fixed-fibre normalizer. The key entry step is a factor-removal lemma that either removes one factor of three at the same tail exponent or produces finite outcome provenance. This prevents an unranked return from a factor-free three/two frame to the preceding exponent-one state. A one-shot initialization then allows the typed normalizer to be entered without treating a larger temporary inner source as a source decrease. The resulting well-founded marked normalization excludes all draw positions.

Contents

1 The game and the theorem

The current state is a positive odd integer n. For $n > 1$ the player to move chooses one of

$$3n - 1, \qquad 3n + 1,$$

and divides the chosen value by the largest possible power of 2, leaving an odd integer. If the resulting odd integer is 1, the player who made the move wins immediately. Players alternate with perfect information.

A position is called WIN for the player to move if at least one move leads to LOSS. It is called LOSS if every legal move leads to WIN. These two classes are the inductively generated finite-remoteness classes. Any position not in either class is called DRAW.

Theorem 1.1 (Main theorem)**.** *Every positive odd starting position has finite remoteness. Equivalently, the $3n \pm 1$ game has no* DRAW *positions, and optimal play reaches* 1 *after finitely many moves.*

The theorem is not a statement about arbitrary play. There are legal cycles, such as

$$5 \longrightarrow 7 \longrightarrow 5,$$

under suitable choices by the players.

2 Binary conjugation

Write an odd state as $n = 2m + 1$ and define

$$F(m) = \left\lceil \frac{3m}{2} \right\rceil.$$

For a nonnegative integer x, let $R(x)$ be the integer obtained by deleting the maximal alternating suffix of the binary expansion of x. Thus $R(1001_2) = 10_2$ and $R(10101_2) = 0$.

Proposition 2.1 (First normal form)**.** *For $m > 0$, the two children in m-coordinates are exactly*

$$F(m), \qquad F(R(m)).$$

Proof. One has

$$3(2m + 1) - 1 = 2(3m + 1), \qquad 3(2m + 1) + 1 = 2(3m + 2).$$

Exactly one of $3m + 1$ and $3m + 2$ is odd. In the other expression, each additional factor of two removed corresponds to crossing one additional alternating bit at the end of the binary expansion of m. The process stops at the first repeated adjacent bit, which is precisely the deletion defining $R(m)$. Returning to m-coordinates gives the displayed pair. □

Since both children lie in the image of F, we represent the state $F(q)$ by q. The conjugated game has terminal state 0 and moves

$$A(q) = \left\lceil \frac{3q}{2} \right\rceil, \qquad B(q) = R(A(q)).$$

Proposition 2.2 (Strict contraction)**.** *For every $q > 0$,*

$$B(q) < q.$$

Proof. Deleting a nonempty binary suffix gives

$$B(q) \le \left\lfloor \frac{A(q)}{2} \right\rfloor \le \left\lfloor \frac{3q+1}{4} \right\rfloor < q$$

for $q \ge 2$, while $B(1) = 0$. □

Thus all unbounded difficulty comes from the expanding branch A and from the unbounded alternating suffix removed by B.

3 Constant-tail coordinates and coefficient sources

For a positive odd integer a, an exponent $r \ge 1$, and a phase $e \in \{0, 1\}$, put

$$Q_r^e(a) = a2^r - e.$$

This is a binary word whose final constant block has length r and consists of the bit e.

Lemma 3.1 (Long-tail recurrence)**.** *For every positive odd a, phase e, and $r \ge 3$,*

$$A(Q_r^e(a)) = Q_{r-1}^e(3a), \qquad B(Q_r^e(a)) = Q_{r-2}^e(3a).$$

Moreover

$$B(Q_1^e(a)) = B(Q_2^e(a)).$$

Proof. For $r \ge 3$, the expanding child still ends in at least two equal bits; hence its maximal alternating suffix consists only of the final bit. This gives both long-tail identities. At the boundary, the binary words of $3a - e$ and $6a - e$ differ by appending one bit that extends the same alternating suffix, so the same prefix remains after deletion. □

Define the embedded three-free coefficient

$$J(s) = 2A(s) + 1.$$

Every positive odd coefficient a has a unique factorization

$$a = 3^k J(s), \qquad k \ge 0.$$

Indeed, remove all factors of 3 and call the remaining odd integer c. Then $c \not\equiv 0 \pmod 3$ and $(c-1)/2$ is 0 or 2 modulo 3. The image of

$$A(2t) = 3t, \qquad A(2t+1) = 3t + 2$$

is exactly the set of nonnegative integers not congruent to 1 modulo 3, and A is injective. Hence there is a unique s with $A(s) = (c-1)/2$, equivalently $c = J(s)$. Unique factorization supplies the unique exponent k. The integer s is called the coefficient source of a and is denoted $\mathrm{src}(a)$.

Lemma 3.2 (Source boundary)**.** *For every $s > 0$ there is a phase*

$$\alpha(s) = 1 - \left(\left\lfloor \frac{s}{2} \right\rfloor \bmod 2 \right)$$

such that

$$\operatorname{odd}(3J(s) + 1 - 2\alpha(s)) = J(A(s))$$

with 2-adic valuation one, while the opposite phase selects $J(B(s))$ with valuation at least two.

Proof. Put $u = A(s)$. Then $J(s) = 2u + 1$ and

$$3J(s) + 1 - 2e = 2(3u + 2 - e).$$

The bracket is odd exactly when $e = 1 - (u \bmod 2)$. For $s = 2t$ or $2t + 1$ one has $u \bmod 2 = t \bmod 2$, so this phase is exactly $e = \alpha(s)$. If u is even, then $e = 1$ and $3u + 1 = J(u)$; if u is odd, then $e = 0$ and $3u + 2 = J(u)$. Thus the valuation-one child is $J(A(s))$. The opposite sign is the other original $3n \pm 1$ child of the odd state $J(s)$; under the conjugacy of Proposition 2.1 that child is $J(B(s))$. Since it is not the valuation-one branch, its valuation is at least two. □

Multiplying a constant-tail coefficient by 3 preserves its coefficient source. Passing through a factor-free signed boundary with coefficient $J(s)$ therefore exposes exactly one ordinary A/B move on the source.

For a positive state q, write $\kappa(q)$ for the odd coefficient in its canonical constant-tail coordinates.

Lemma 3.3 (Coefficient pullback)**.** *For every positive odd w,*

$$\kappa(3w) = J(R(w)), \qquad \kappa(3w - 1) = J(R(2w - 1)).$$

Proof. Let $m \geq 1$ be the length of the maximal alternating suffix of the odd word w, and put $z = R(w)$. If $z > 0$, maximality says that the leading bit of the suffix equals the last bit of z. Because the suffix ends in 1, its value is

$$t_m = \begin{cases} (2^m - 1)/3, & m \text{ even}, \\ (2^{m+1} - 1)/3, & m \text{ odd}. \end{cases}$$

Thus $w = 2^m z + t_m$. If m is even, z is even and

$$3w + 1 = 2^m(3z + 1) = 2^m J(z).$$

If m is odd, z is odd and

$$3w + 1 = 2^m(3z + 2) = 2^m J(z).$$

If $z = 0$, the whole word is alternating; then m is odd and $3w + 1 = 2^{m+1}$, whose odd part is $J(0) = 1$. Hence in all cases $\kappa(3w) = \operatorname{odd}(3w + 1) = J(R(w))$.

For the second identity set $v = 2w - 1$, which is odd. Since $3v + 1 = 2(3w - 1)$, the first identity applied to v gives

$$\kappa(3w - 1) = \operatorname{odd}(3w - 1) = \operatorname{odd}(3v + 1) = J(R(v)) = J(R(2w - 1)).$$

□

4 Finite outcomes, witnesses, and ranks

The terminal state 0 is LOSS with height $h(0) = 0$. A finite WIN state has height

$$h(x) = 1 + \min\{h(y) : y \text{ is a LOSS child of } x\},$$

and a finite LOSS state has height

$$h(x) = 1 + \max\{h(y) : y \text{ is a child of } x\}.$$

A state whose outcome has been proved but which is being carried only to justify a later incidence is called a *routing witness.* A *retained proof token* is a finite state for which the proof has also established the comparison needed by the rank: it either seeds an empty inner token set at a one-shot seed transition, or replaces a previously retained token by certified descendants of strictly smaller height. This distinction is important: a newly discovered finite state is never silently appended to a ranked multiset.

Lemma 4.1 (Multiset token rank)**.** *For a finite multiset M of proof heights define*

$$\Phi(M) = \mathop{\#}_{h \in M} \omega^h,$$

the natural ordinal sum of the monomials ω^h. Replacing one token of height H by any finite multiset of certified descendants of heights strictly below H strictly decreases Φ.

Proof. In Cantor normal form the coefficient of ω^H decreases by one, while all inserted terms have lower exponent. No finite collection of lower terms can compensate for the loss at exponent H. □

Every positive state q has unique canonical constant-tail coordinates $q = Q_r^e(a)$ with a odd and $r \geq 1$. If $a = 3^k J(s)$, define the *state source* $\rho(q) = s$.

Lemma 4.2 (Universal state-source bound)**.** *For every $q > 0$,*

$$\rho(q) \leq \frac{q-1}{6}.$$

Proof. Write $q = Q_r^e(3^k J(s))$. Since $r \geq 1$ and $e \leq 1$, $q \geq 2J(s) - 1$. Moreover $J(s) = 2\lceil 3s/2 \rceil + 1 \geq 3s + 1$. Hence $q \geq 6s + 1$. □

Lemma 4.3 (Boundary coefficient descent)**.** *Let a be positive and odd and*

$$p = B(Q_1^e(a)) = B(Q_2^e(a)).$$

If $p > 0$, then its canonical coefficient satisfies

$$\kappa(p) \leq \frac{3a+1}{4}.$$

In particular $\kappa(p) < a$ for $a \geq 3$.

Proof. The common child is $p = R(3a - e)$. Deleting a nonempty suffix gives $p \leq \lfloor (3a-e)/2 \rfloor$. For every positive integer y, the odd coefficient in its canonical constant-tail coordinates is at most $(y+1)/2$. Therefore

$$\kappa(p) \leq \frac{p+1}{2} \leq \frac{3a+1}{4}.$$

The final inequality is strict relative to a for $a \geq 3$. □

5 Side identities and typed boundary objects

The finite routing system uses two elementary binary identities. We record them here because they are the only places where an exceptional residue class enters.

Lemma 5.1 (Ordinary side relation). *Put $S(q) = B(q)$ and $T(q) = B(A(q))$. If*

$$q \not\equiv 1, 3, 12, 14 \pmod{16},$$

then

$$T(q) \in \{A(S(q)), B(S(q))\}.$$

If q is in one of the four exceptional classes, then $A(q)$ is not exceptional.

Proof. Let $x = A(q)$ and let k be the length of the maximal alternating suffix of x. Direct reduction of $A(q)$ modulo 8 shows that $k \geq 3$ is possible exactly for $q \equiv 1, 3, 12, 14 \pmod{16}$. Outside these classes $k = 1$ or 2. Writing x as $S(q)$ followed by that one- or two-bit alternating word, the formulas

$$A(2u) = 3u, \qquad A(2u + 1) = 3u + 2$$

show that $A(x)$ is obtained from $A(S(q))$ by appending the corresponding one- or two-bit alternating word. Deleting its maximal alternating suffix therefore leaves either $A(S(q))$ or $B(S(q))$. The four exceptional residue classes map under A to $2, 5, 10, 13 \pmod{16}$, none of which is exceptional. □

Corollary 5.2 (No four consecutive wins). *For no q can all four states*

$$q, A(q), A^2(q), A^3(q)$$

be WIN.

Proof. If q and $A(q)$ are both WIN, then $B(q)$ must be LOSS; the same holds at the next two positions. If q is ordinary, Lemma 5.1 makes $B(A(q))$ a child of the LOSS state $B(q)$, impossible. Hence q must be exceptional. Applying the same argument to $A(q)$ forces $A(q)$ exceptional, contrary to Lemma 5.1. □

Lemma 5.3 (Alternating lift identity). *For every positive integer z there are $m \geq 1$ and $\delta = 1 - (z \bmod 2)$ such that*

$$3z + 1 = Q_m^{\delta}(J(R(z))).$$

If $R(z) > 0$, one may take m equal to the length of the maximal alternating suffix of z; if $R(z) = 0$, one takes one more than the binary length of z.

Proof. First suppose $r = R(z) > 0$, let k be the alternating-suffix length and put $\epsilon = r \bmod 2$. Then

$$z = 2^k r + t_{k,\epsilon}, \qquad t_{k,0} = \left\lfloor \frac{2^k}{3} \right\rfloor, \quad t_{k,1} = \left\lfloor \frac{2^{k+1}}{3} \right\rfloor.$$

Since $J(r) = 3r + 1 + \epsilon$, direct substitution gives

$$3t_{k,\epsilon} + 1 + \delta = 2^k(1 + \epsilon),$$

the two parity cases being exactly the two remainders of a power of 2 modulo 3. Hence

$$3z + 1 + \delta = 2^k J(r),$$

which is the required identity with $m = k$. If $R(z) = 0$, the whole binary word of z is alternating and starts with 1, so $z = t_{k,1}$. The same calculation gives $3z + 1 + \delta = 2^{k+1} = 2^{k+1}J(0)$. □

Lemma 5.4 (Exceptional side attachment)**.** *Let q be exceptional and suppose*

$$q \text{ is WIN}, \qquad A(q) \text{ is DRAW}, \qquad \ell = B(q) \text{ is LOSS}.$$

Then for some $m \geq 1$ and $\delta \in \{0, 1\}$ the two children of $A(q)$ are exactly

$$Q_m^\delta(J(\ell)), \qquad Q_{m+1}^\delta(J(\ell)),$$

and this adjacent pair contains a continuing DRAW.

Proof. Write $q = 16t + c$ with $c \in \{1, 3, 12, 14\}$. The four direct exceptional formulas are

c	ℓ	$B(A(q))$	$A^2(q)$
1	$R(6t)$	$18t + 1$	$36t + 3$
3	$R(6t + 1)$	$18t + 4$	$36t + 8$
12	$R(6t + 4)$	$18t + 13$	$36t + 27$
14	$R(6t + 5)$	$18t + 16$	$36t + 32.$

In every row put respectively $z = 6t, 6t+1, 6t+4, 6t+5$. Then $\ell = R(z)$ and $B(A(q)) = 3z+1$. By Lemma 5.3,

$$B(A(q)) = Q_m^\delta(J(\ell))$$

for some $m \geq 1$. The last column is $2B(A(q)) + \delta$, so it equals $Q_{m+1}^\delta(J(\ell))$. These are precisely the two children of the DRAW state $A(q)$; therefore neither is LOSS and at least one is DRAW. □

For a source x and phase e, let $O(x, e)$ denote the following exact *DRAW obligation*:

(a) $Q_1^e(J(x))$ is DRAW; or

(b) there is a DRAW state whose two children are exactly $Q_2^e(J(x))$ and $Q_1^e(J(x))$.

This definition introduces no outcome assumption beyond the displayed DRAW state.

Lemma 5.5 (Hidden parent for a typed three/two frame)**.** *Let a be positive, odd and not divisible by 3, and suppose*

$$a \equiv 1 + e \pmod 3.$$

Put $H = Q_3^e(a)$, $G = Q_2^e(a)$ and

$$K = \frac{2H + e - 1}{3}.$$

Then K is an integer and

$$\text{moves}(K) = \{H, G\}.$$

Proof. For $e = 0$ the congruence gives $H \equiv 2 \pmod 3$ and $K = (2H - 1)/3$; for $e = 1$ it gives $H \equiv 0 \pmod 3$ and $K = 2H/3$. These are exactly the two inverse branches of the injective map $A(q) = \lceil 3q/2 \rceil$. The last three bits of H are equal, so the contracting child removes only the last bit and equals G. □

6 Fixed-tail factor removal and entry typing

The next two lemmas isolate the only raw entry phenomenon that is not already an obligation or a factor frame.

Lemma 6.1 (Fixed-tail factor-removal diamond)**.** *Let c be positive and odd, $e \in \{0,1\}$, and $r \geq 2$. Put*

$$X = Q_r^e(3c), \qquad G = Q_r^e(c), \qquad H = Q_{r+1}^e(c), \qquad Y = A(G).$$

Then

$$\mathrm{moves}(H) = \{X, Y\}.$$

If X is DRAW*, either G is* DRAW*, or the diamond determines a finite routing witness together with its exact incidence to G, H, X, Y and the other child of G.*

Proof. Since $r+1 \geq 3$, Lemma 3.1 gives $A(H) = X$. For $r \geq 3$ it also gives $B(H) = Q_{r-1}^e(3c) = A(G)$; for $r = 2$ the boundary identity gives the same equality. Thus $B(H) = Y$.

Because X is DRAW, H is not LOSS. If H is WIN, then Y is a LOSS child of H. Assume H is DRAW. Then Y is not LOSS. If G is LOSS, then Y is WIN. If G is WIN, either Y itself is a LOSS witness, or the other child of G is a LOSS witness. These are all outcome orientations. In every finite branch the witness and the parent-child incidence just displayed are known exactly; no token-rank comparison is asserted at this stage. □

Definition 6.2 (Entry controls)**.** *A* boundary entry *is an obligation $O(x, e)$ or, more generally, a factor-free adjacent frame*

$$\{Q_{m+1}^e(J(x)), Q_m^e(J(x))\}, \qquad m \geq 1,$$

known to contain a continuing DRAW*. Its remaining tail length is recorded as τ. The typed three/two frame satisfying Lemma 5.5 is a distinguished $m = 2$ boundary constructor. A* loss-sibling entry *is one of the finitely many local constructors in which a continuing* DRAW *is accompanied by a finite* WIN/LOSS *routing witness with a proved parent/child incidence: the fixed-tail diamond of Lemma 6.1, the exponent-one predecessor diamond of Lemma 6.4, an ordinary side diamond of Lemma 5.1, the exceptional side attachment of Lemma 5.4, or the hidden-parent frame of Lemma 5.5. The constructor records its remaining constant-tail/factor exponent as the local counter τ.*

Lemma 6.3 (Entry typing)**.** *Every finite branch produced by Lemma 6.1 is a loss-sibling entry in the sense of Definition 6.2; no ranked multiset is changed in order to make this classification.*

Proof. The proof of Lemma 6.1 gives the finite witness and its exact incidence to G, H, X, Y and the other child of G. These are precisely the data of the first loss-sibling constructor. Classification is therefore immediate; subsequent tail/factor normalization belongs to the typed routing relation and is ranked there by τ and, when available, by descendant-token comparisons. □

Lemma 6.4 (Exponent-one lift)**.** *Let a be positive and odd and $k \geq 1$. If $R_k = Q_1^e(3^k a)$ is* DRAW*, then*

$$P_{k-1} = Q_2^e(3^{k-1} a)$$

is either DRAW *or* WIN*; in the latter case its other child is a finite* LOSS *witness.*

Proof. The boundary recurrence gives $A(P_{k-1}) = R_k$. A state with a DRAW child cannot be LOSS. If P_{k-1} is WIN, the DRAW child cannot witness the win, so its other child is LOSS. □

Proposition 6.5 (Arbitrary factorful exponent-one entry)**.** *Let $a = J(s)$ and $k > 0$. If $Q_1^e(3^k a)$ is* DRAW*, then after finitely many outcome-compatible local steps one obtains either a typed entry of Lemma 6.3, or the factor-free exponent-two state*

$$Q_2^e(a)\ \text{DRAW}.$$

Proof. Apply Lemma 6.4. Its finite branch is a loss-sibling entry by Definition 6.2. Otherwise $Q_2^e(3^{k-1}a)$ is DRAW. Apply Lemma 6.1 with $r = 2$ repeatedly. Whenever its finite branch occurs, Lemma 6.3 supplies a typed entry. If it never occurs, one factor of 3 is removed at the same tail exponent at each step, so after $k - 1$ steps the displayed factor-free exponent-two state is reached. □

Proposition 6.6 (Complete exponent-one/two boundary reduction)**.** *Let $s \geq 0$, $k \geq 0$, $e \in \{0, 1\}$, and let*

$$Q_r^e(3^k J(s)), \qquad r \in \{1, 2\},$$

be DRAW*. For $s > 0$, after finitely many local steps at least one of the following occurs: a typed entry is obtained, the retained source strictly decreases, or the factor-free exponent-two state $Q_2^e(J(s))$ is* DRAW*. For $s = 0$ the corresponding statement is Lemma 8.1.*

Proof. Assume $s > 0$. There are four cases. If $r = 1, k = 0$, the state is the first alternative of the obligation $O(s, e)$ and is already a boundary entry. If $r = 1, k > 0$, use Proposition 6.5. If $r = 2, k = 0$, we are at the displayed factor-free exponent-two state. Finally, if $r = 2, k > 0$, apply Lemma 6.1 repeatedly with $r = 2$: a finite branch is a loss-sibling entry by Definition 6.2, while otherwise one factor of 3 is removed at the same exponent at each step and after k steps the factor-free exponent-two state is reached. These cases are disjoint and exhaustive. □

7 Factor-free exponent-two base entry

Let $x > 0$, $a = J(x)$ and

$$P = Q_1^e(a), \quad U = Q_2^e(a), \quad b = B(P) = B(U), \quad F = A(U) = Q_1^e(3a), \quad g = 1 - e.$$

Assume U is DRAW and x is the least coefficient source of any DRAW. The common child b cannot be 0, because 0 is LOSS and a DRAW has no LOSS child. By Lemma 4.3, b has canonical coefficient strictly below $J(x)$; because J is strictly increasing, its state source is below x. Hence b is not DRAW. As a child of a DRAW it is not LOSS, so

$$b \text{ is WIN}, \qquad F \text{ is DRAW}.$$

Lemma 7.1 (First-factor anchor identity)**.** *For the states above,*

$$9J(x) + 1 - 2e = 2^v J(b), \qquad v \geq 1. \tag{1}$$

Proof. Put $a = J(x)$. If $e = 0$, then $P = 2a$, $A(P) = 3a$ and $b = R(3a)$. Apply the first identity of Lemma 3.3 to the odd integer $w = 3a$:

$$\text{odd}(9a + 1) = \kappa(9a) = J(R(3a)) = J(b).$$

If $e = 1$, then $P = 2a - 1$, $A(P) = 3a - 1$ and $b = R(3a - 1)$. The second identity of Lemma 3.3, again with $w = 3a$, gives

$$\mathrm{odd}(9a - 1) = J(R(6a - 1)).$$

By the boundary identity of Lemma 3.1, $R(6a - 1) = R(3a - 1) = b$. Hence in both phases the odd part of $9a + 1 - 2e$ is exactly $J(b)$, which proves (1). ☐

Proposition 7.2 (Base-entry attachment)**.** *Every factor-free exponent-two* DRAW *at a globally minimal source has a finite continuation which either strictly lowers the retained source or produces a boundary/loss-sibling typed entry.*

Proof. Suppose first that $e = 1 - \alpha(x)$. The first source-boundary numerator is divisible by 4, while

$$9J(x) + 1 - 2e = 3(3J(x) + 1 - 2e) - 2(1 - 2e) \equiv 2 \pmod 4.$$

Thus $v = 1$. Writing $u = A(x)$ gives $b = 3u + 1$. The two children of the DRAW state F are

$$T = Q_1^g(J(b)), \qquad C = R(T),$$

and direct appended-bit deletion gives $C \in \{A(b), B(b)\}$. If T is DRAW, this is the boundary entry $O(b, g)$. If C is DRAW, the other child ℓ of the WIN state b is a finite LOSS witness. If b is ordinary, this is the ordinary-side loss-sibling constructor. If b is exceptional and $C = A(b)$, Lemma 5.4 gives an adjacent DRAW frame over ℓ, hence a boundary/loss-sibling entry. It remains only the exceptional orientation $C = B(b)$. In that orientation $b \le (9x + 5)/2$ and the alternating suffix of $A(b)$ has length at least three, so

$$C = B(b) \le \frac{A(b)}{8} \le \frac{3b + 1}{16}.$$

Lemma 4.2 then yields

$$\rho(C) \le \frac{C - 1}{6} < \frac{3b + 1}{96} \le \frac{27x + 17}{192} < x,$$

a strict retained-source exit.

Now suppose $e = \alpha(x)$. Lemma 3.2 gives $3J(x) + 1 - 2e = 2J(A(x))$, hence $v \ge 2$. If $v \ge 4$, then

$$16J(b) \le 9J(x) + 1 \le 27x + 19,$$

and $J(b) \ge 3b + 1$ implies

$$b \le \frac{9x + 1}{16} < x.$$

Thus the continuing DRAW has smaller retained source. If $v = 2$, the two children of F are exactly $Q_2^g(J(b))$ and $Q_1^g(J(b))$, which is the second form of $O(b, g)$. If $v = 3$, they are the typed three/two frame $Q_3^g(J(b)), Q_2^g(J(b))$. Reducing (1) modulo 3 gives

$$J(b) \equiv 1 + g \pmod 3,$$

so Lemma 5.5 applies. The hidden parent is a boundary entry; its finite side, if one is exposed, is a loss-sibling entry. Hence no untyped base row remains. ☐

8 The source-zero branch

For $k \geq 0$, $r \geq 1$ and $e \in \{0,1\}$ put

$$S^e_{k,r} = Q^e_r(3^k).$$

These are precisely the constant-tail states with coefficient source zero.

Lemma 8.1 (Zero-source boundary normalization)**.** *If a source-zero* DRAW *exists, then after finitely many outcome-compatible moves one reaches a typed boundary or loss-sibling entry. The one-shot entry bit is not consumed before this finite normalization is complete.*

Proof. For $r \geq 3$, Lemma 3.1 gives

$$\operatorname{moves}(S^e_{k,r}) = \{S^e_{k+1,r-1}, S^e_{k+1,r-2}\}.$$

Following a DRAW child strictly lowers r, so a DRAW with $r \in \{1,2\}$ is reached after finitely many moves. If $k = 0$ already at that boundary, the four possibilities are explicit:

$$Q^1_1(1) = 1 \text{ WIN}, \quad Q^1_2(1) = 3 \text{ WIN}, \quad Q^0_1(1) = 2 \text{ LOSS}, \quad Q^0_2(1) = 4 \text{ LOSS}.$$

Hence a boundary DRAW must have accumulated at least one factor of 3.

At the boundary write, for some $n \geq 1$,

$$3^n + 1 - 2e = 2^j J(y).$$

The exact 2-adic valuation is

	n odd	n even
$e = 0$	$j = 2$	$j = 1$
$e = 1$	$j = 1$	$j = 2 + v_2(n)$.

For $j \geq 2$ the raw and signed exits form the adjacent factor-free frame

$$Q^{1-e}_{j-1}(J(y)), \qquad Q^{1-e}_j(J(y)),$$

containing a continuing DRAW; by Definition 6.2 this is a boundary entry, with its finite tail length recorded in τ. For $j = 1$ one has

$$y = \frac{3^{n-1} - 1}{2}.$$

When $e = 0$ the raw exit is $A(y)$; when $e = 1$ it is $B(y)$. The raw and signed states are the two exits of the same boundary DRAW configuration. If the signed exit is DRAW, it is already the exponent-one form of an obligation. If the raw exit is the continuing DRAW and the signed exit is finite, that finite state is necessarily WIN; choosing a canonical LOSS child of it supplies an exact finite routing witness, hence a loss-sibling entry. If both exits are DRAW, use the signed obligation. Thus every row reaches a typed entry after a finite pre-entry computation; no impossible "source below zero" comparison is used. ☐

Lemma 8.2 (A-selecting side diamond)**.** *Let $x > 0$, $e = \alpha(x)$, $w = Q^e_1(J(x))$, and $y = A(x)$. Then*

$$B(w) \in \{A(y), B(y)\}.$$

Proof. The source-boundary identity gives $A(w) = Q^{1-e}_1(J(y)) = 4A(y) + 1 + e$. For $e = 0$ the appended bits are 01; for $e = 1$ they are 10. If the first appended bit repeats the last bit of $A(y)$, deleting the alternating suffix leaves $A(y)$; otherwise the deletion continues through the maximal alternating suffix of $A(y)$ and leaves $B(y)$. ☐

Lemma 8.3 (Universal B-transfer diamond). *Let $x > 0$, let $e = 1 - \alpha(x)$, put $a = J(x)$ and $g = 1 - e$, and factor*

$$3a + 1 - 2e = 2^j J(y), \qquad j \geq 2,$$

so $y = B(x)$. Define

$$P = Q_1^e(a), \quad U = Q_2^e(a), \quad D = Q_j^g(J(y)), \quad C = Q_{j-1}^g(J(y)), \quad F = A(U).$$

Then $D = A(P)$ and $C = B(P) = B(U)$. If X is the common child of D, C and Y is the other child of C, then

$$B(F) = Y.$$

Consequently an obligation in the B-selecting phase transfers a continuing DRAW *to the adjacent frame $\{D, C\}$; when $j = 2$ the transferred frame is again an obligation in the opposite phase.*

Proof. The formulas for D and C are the signed boundary formula and the shared-child identity. The remaining equality is a direct substitution in the three cases $j = 2$, $j = 3$, and $j \geq 4$: the constant suffix of D determines its common child X, while the one-step longer factor state F has the same raw side Y. For the outcome assertion, a common child of a DRAW cannot be LOSS. If the exponent-one member is DRAW, the frame $\{D, C\}$ already contains a DRAW. In the parent form of an obligation, the only remaining orientation has the exponent-two member DRAW; if C were WIN, outcome recursion through the displayed diamond would force Y both LOSS and a child of the DRAW state F, impossible. Hence C is DRAW. For $j = 2$, D, C are exactly the exponent-two/one pair over $J(y)$. $\square$

9 The typed routing system

We now define the marked relation used after entry. A configuration contains an outer retained source s, an outer token multiset M, an entry bit $\eta \in \{1, 0\}$, and, after entry, inner data

$$(a, \zeta, N, v, \tau).$$

Here a is the retained inner source anchor; N is the finite multiset of inner proof-token heights; $\zeta \in \{1, 0\}$ is a one-shot token-seed bit; v is one of the finite control states below; and $\tau \in \mathbb{N}$ is a local tail/factor counter. Temporary arithmetic cursors are not rank anchors. An inner source is promoted to the ranked anchor a only when every already retained token remains a valid witness for the new typed state; otherwise the numerical value remains a cursor until a preceding token decrease permits a reset.

The bit $\eta = 1$ means that the current retained-source fibre has not yet initialized the typed relation. The strict transition $\eta : 1 \to 0$ may install the initial finite outer-token multiset M, an arbitrary finite inner routing source a, and any already proved routing witnesses. This is why η is placed *before* the token and inner-source components of the rank. After $\eta = 0$, M can change only by certified strict descendant replacement. If entry contains no token suitable for ranking, the inner seed bit is set to $\zeta = 1$; the first later certified token may be installed in N by the strict transition $\zeta : 1 \to 0$. Once $\zeta = 0$, every change of N is a certified strict descendant replacement.

Define

$$\Psi(N) = \mathop{\#}_{h \in N} \omega^h.$$

The full lexicographic rank is

$$\mathcal{R} = (s, \eta, \Phi(M), \zeta, \Psi(N), a, c(v), \tau). \tag{2}$$

Thus an entry-bit decrease may seed M and a; a later token decrease may reset the still-later source/control data. All natural components use their usual well-order and both bits are ordered $1 > 0$.

The finite control set $\mathcal{V}$ consists of the fifteen names listed in Table 1; the names are labels for semantic constructors, not additional mathematical assumptions.

Lemma 9.1 (A-obligation routing). *An A-selecting obligation $O(x, \alpha(x))$ has only the following outcome-compatible exits: a strict inner-source decrease, a strict inner-token replacement, a bounded canonical A-streak, or a factor fork carrying an actual finite witness.*

Proof. Let $e = \alpha(x)$, $g = 1 - e$, $P = Q_1^{\mathrm{e}}(J(x))$, $U = Q_2^{\mathrm{e}}(J(x))$ and $b = B(P) = B(U)$. Lemma 8.2 makes b an ordinary child of $A(x)$ and gives

$$\rho(b) \le \frac{9x+1}{24} < x.$$

Thus a DRAW at b is a strict inner-source exit. Suppose b is WIN. If P is DRAW, its other child is DRAW and is exactly the next obligation $O(A(x), 1-e)$. In the parent form of $O(x, e)$, the exponent-one state is WIN, the exponent-two state is DRAW, and outcome recursion forces an actual LOSS sibling; the other child of the exponent-two state is an adjacent factor frame carrying that witness. These are the stated rows. ☐

Lemma 9.2 (Bounded canonical A-streak). *A canonical A-continuation from an A-selecting obligation performs at most four equal-rank side tests before reaching a B-selecting control or a strict source/token exit.*

Proof. During the streak the side states are consecutive points x_2, x_3, x_4, x_5 on one A-ray. If all four were WIN, this would contradict Corollary 5.2. The first non-WIN side state is either a DRAW, giving a strict source exit in the ordinary case or the adjacent frame of Lemma 5.4 in the exceptional case, or the phase has already become B-selecting. Thus no fifth equal-rank test exists. ☐

Lemma 9.3 (B-selecting transfer). *Every `b_select` control carries, as part of its provenance, a retained anchor p and a current B-selecting cursor y with*

$$y \le \frac{3p+1}{2}.$$

Let $j \ge 2$ be the selected signed valuation and $z = B(y)$. If $j \ge 3$, then

$$z \le \frac{27p+7}{48} < p,$$

so the retained inner source strictly decreases. If $j = 2$, the transferred frame is again an exact obligation. Its first occurrence may use the one-shot token seed; every later compatible valuation-two recycle either strictly lowers the retained inner source or replaces the retained witness by a certified lower descendant.

Proof. Lemma 8.3 sends the B-selecting obligation to an adjacent frame over $z = B(y)$. When $j \ge 3$, at least two additional suffix bits are deleted after the mandatory first division, and the exact source estimate is

$$z \le \frac{9y-1}{24} \le \frac{27p+7}{48} < p.$$

For $j = 2$ the transferred exponent-two/one pair is exactly the same obligation type in the opposite phase. The first occurrence exposes the finite ordinary-side witness and, if no inner token has yet been seeded, uses $\zeta : 1 \to 0$. In a later valuation-two recycle the common-side identity places the new witness below the currently retained finite witness in its proof tree; hence its height is strictly smaller. No unrelated token is inserted after $\zeta = 0$. ☐

Lemma 9.4 (Boundary-entry normalization)**.** *A* `boundary_entry` *either makes a strict source/token transition or, after finitely many equal-rank tail steps, reaches* `a_obligation` *or* `factor_fork`*.*

Proof. For an obligation there is nothing to normalize. Otherwise the entry is an adjacent factor-free frame with finite lower exponent m. Follow a continuing DRAW through the long-tail recurrence while the lower exponent is at least three. Each pure routing step lowers the recorded tail counter τ. At the first exponent-one/two boundary, the shared-child/source identities give an obligation if the coefficient is factor-free and a factor fork if powers of 3 have accumulated. The hidden-parent congruence handles the typed three/two constructor without introducing an additional row. Thus the macro is finite; any earlier source or certified token exit is already strict. □

Lemma 9.5 (Loss-sibling normalization)**.** *A* `loss_sibling_entry` *either makes a strict token/source transition or reaches* `factor_fork` *after finitely many equal-rank steps.*

Proof. The entry records one of the finite constructors of Definition 6.2 and the remaining tail/factor length τ. If the constructor is already an exponent-one/two or hidden-parent frame, it is a factor fork. Otherwise, choose the continuing DRAW branch in the long-tail diamond. Lemma 3.1 reduces the remaining tail length. Whenever the carried finite witness meets the continuing branch in an ordinary side diamond, Lemma 5.1 either keeps the same witness incidence or replaces it by an actual child, which is a strict proof-height replacement. An exceptional side can occur for only one step because its A-successor is ordinary. Thus an equal-rank passage strictly decreases τ and must reach the exponent-one/two factor boundary. No new witness is introduced during this countdown. □

Lemma 9.6 (Factor and marked-tail routing)**.** *Every factor-fork, high-return, marked-tail, short-lift, and terminal control has one of the following effects:*

(a) strict inner-source decrease;

(b) strict replacement of an existing inner token by certified lower descendant token(s);

(c) a control change listed in Table 1 with the retained source and token multiset unchanged; or

(d) a same-control recurrence with strictly smaller local counter τ.

Proof. The boundary and loss-sibling controls are covered by Lemmas 9.4 and 9.5.

The factor exponent-one row uses the shared exponent-one/two child. If the selected finite source is ordinary, Lemma 5.1 exposes a common child of the retained LOSS witness and hence a lower WIN token; the exceptional row is an exact constant-tail lift over the same LOSS witness. For factor exponent at least two, the reverse constant-tail parent either removes a factor at exponent at least two or exposes a finite parent-child witness; in the latter case its common child is a certified descendant.

A high return has valuation $5, 6$, or at least 7. In the first two rows the explicit short-tail formulas replace the carried return token before the new marked task is installed. For valuation at least 7, the universal three/two frame carries a WIN/LOSS pair; the selected common child is a certified descendant before the tail is restarted. Hence every high-return edge is strict in $\Psi(N)$.

For a marked constant tail of length $D \geq 3$, if its marked states are q (WIN) and y (LOSS), the common child

$$r = B(q) = A(y)$$

for $D \geq 4$ (and the boundary common child for $D = 3$) is a WIN child of y, so $h(r) < h(y)$. Thus the existing token y is replaced before any new inner task is installed. The two short rows $D = 1, 2$ are exact lifts over an already carried token and merely change control. Long exact-lift recurrences strictly lower their remaining tail/factor exponent, which is τ. Terminal exponent-two/three macros reconstruct a finite descendant token before they restart any obligation. The strict token replacements used in these rows are collected explicitly in Table 2. These cases exhaust the factor/tail constructors because their only unbounded parameters are the valuation and tail intervals just split above. $\square$

The equal-retained-rank control edges and a concrete control height $c(v)$ are shown in Table 1. Any edge not listed there is strict in an earlier component of (2). Same-control tail/factor recurrences strictly decrease τ.

Control v	$c(v)$	Equal-rank successors
`marked_tail`	9	`short_lift`
`boundary_entry`	8	`a_obligation`, `factor_fork`
`short_lift`	8	`a_obligation`
`a_obligation`	7	`a_test_4`, `factor_fork`
`a_test_4`	6	`a_test_3`, `b_select`
`a_test_3`	5	`a_test_2`, `b_select`
`a_test_2`	4	`a_test_1`, `b_select`
`a_test_1`	3	`b_select`
`b_select`	2	`b2_first`
`loss_sibling_entry`	2	`factor_fork`
`b2_first`	1	`b2_ready`
`factor_fork`	1	`high_return`
`b2_ready`	0	–
`high_return`	0	–
`terminal_macro`	0	–

Table 1: Equal-rank control graph. Every listed edge strictly lowers $c(v)$. Strict source/token/bit edges may reset c and τ.

Proposition 9.7 (Well-founded typed normalizer)**.** *The typed routing relation is well-founded for every typed entry.*

Proof. Consider the rank (2). A retained-source decrease is strict in the first component and may reset all later data. At fixed retained source, $\eta : 1 \to 0$ is strict and may seed M and all later routing data. Once $\eta = 0$, every change of M is a certified strict descendant replacement and therefore decreases $\Phi(M)$ by Lemma 4.1; it may reset all later inner data.

At fixed earlier components, $\zeta : 1 \to 0$ is a one-time strict inner-token seed. Thereafter every change of N strictly decreases $\Psi(N)$. With the token components fixed, a promoted inner-source comparison strictly lowers a. If all preceding components are unchanged, the semantic routing lemmas above show that a control-changing equal-rank edge is one of the edges of Table 1, hence strictly lowers $c(v)$; a same-control tail/factor recurrence strictly lowers τ. Thus every transition strictly decreases one component while preserving all earlier components. The lexicographic product of these well-orders is well-founded. $\square$

10 Proof of the main theorem

Proof of Theorem 1.1. Assume that a DRAW exists and choose the least state source s occurring among DRAW positions. Begin with $\eta = 1$; no ranked token has yet been installed.

If $s = 0$, Lemma 8.1 performs a finite pre-entry normalization and reaches a boundary or loss-sibling typed entry. Consume $\eta : 1 \to 0$ exactly there and invoke Proposition 9.7.

Let $s > 0$. Follow an actual DRAW child through long constant tails until the tail exponent is one or two. Proposition 6.6 then gives a typed entry, a strict retained-source exit, or the factor-free exponent-two state $Q_2^{e}(J(s))$ DRAW. The last case is handled by Proposition 7.2. A strict smaller-source exit contradicts the choice of s; otherwise a typed entry is obtained and η is consumed. Thus all four boundary combinations $(r, k) = (1, 0), (1, > 0), (2, 0)$ and $(2, > 0)$ are covered.

From then on every macro transition lies in the well-founded relation of Proposition 9.7, unless an earlier outer component strictly decreases, in which case the lexicographic rank decreases even more strongly. Yet every DRAW has no LOSS child and at least one DRAW child. Therefore from every marked DRAW configuration an outcome-compatible continuing DRAW can be chosen, yielding another finite macro transition. Iteration would create an infinite descending chain in the rank (2), impossible. Hence no DRAW exists in the conjugated game.

It remains only to transfer the conclusion back to an arbitrary original odd starting state. Write it as $n = 2m + 1$. If $m = 0$, then $n = 1$ is terminal. For $m > 0$, Proposition 2.1 says that its two original children are $F(m)$ and $F(R(m))$. These are represented in the conjugated game by the states m and $R(m)$, respectively, and both have finite remoteness by the no-DRAW conclusion just proved. Hence the original state n is itself WIN or LOSS with finite height. Thus every positive odd starting position has finite remoteness and optimal play reaches 1. $\square$

11 Verification boundary and reproducibility

The theorem above is a human mathematical claim. The accompanying files have narrower roles.

- `verify_v3_0.py` checks the long-tail identities, fixed-tail diamonds, ordinary side relation, base-entry valuations and inequalities, source-zero valuation table, hidden-parent identity, and the finite equal-rank control graph over large finite ranges.
- `routing_certificate_v3_0.json` records the control states, equal-rank edges, strict edge classes, and rank components used in Table 1. It is a structural certificate, not a substitute for the semantic proofs in this article.
- The repository's Lean development checks selected definitions and general well-founded metatheory. It is not claimed to kernel-check the complete concrete theorem.

The finite computations support the arithmetic and the transcription of the routing table; the infinite conclusion rests on the proved identities, outcome recursion, and the well-founded rank.

A Exceptional side formulas

For completeness, the four exceptional classes of Lemma 5.1 have exact formulas. For $t \geq 0$,

q	$B(q)$	$B(A(q))$
$16t + 1$	$R(6t)$	$18t + 1$
$16t + 3$	$R(6t + 1)$	$18t + 4$
$16t + 12$	$R(6t + 4)$	$18t + 13$
$16t + 14$	$R(6t + 5)$	$18t + 16.$

The displayed $B(A(q))$ value always has constant-tail length one. Hence an exceptional side step cannot repeat immediately; its next A-source is ordinary. These formulas also justify the exceptional branch used in Proposition 7.2.

B Routing inventory

The equal-rank graph in Table 1 is deliberately smaller than the full outcome case split: source and token exits are omitted from the graph because they are already strict in earlier components. The semantic partitions are as follows.

Control	Exhaustive nonterminal partition
`boundary_entry`	source/token exit, `a_obligation`, or `factor_fork` after finite tail countdown
`loss_sibling_entry`	source/token exit, or `factor_fork` after finite tail countdown
`a_obligation`	source exit, token exit, canonical A test, or retained-witness factor fork
`a_test_4,3,2,1`	source exit, token exit, next test, or B-selecting switch; the last test has no next-test row
`b_select`	valuation 2, or valuation at least 3 (strict source)
`b2_first`	strict source, or one-time witness installation
`b2_ready`	strict source, or strict witness descendant
`factor_fork`	exponent-one source/token, higher source/token, or adjacent high return
`high_return`	valuations $5, 6$ or at least 7; each spends a token before marked-tail restart
`marked_tail`	$D = 1$, $D = 2$, or $D \geq 3$; long row replaces the marked LOSS token by its common WIN child
`short_lift`	exact lift into an obligation
`terminal_macro`	exponent-two/three terminal rows; each begins after a strict token replacement

This inventory is exactly the case structure used in the proof of Proposition 9.7; there is no implicit "other" row.

Routing row	Retained replacement	Certified comparison
ordinary factor fork	$b \mapsto X$	X is a child of the canonical LOSS child of b, hence $h(X) \leq h(b) - 2$
exceptional factor fork	$b \mapsto \ell$	ℓ is the actual LOSS child of b, hence $h(\ell) = h(b) - 1$
high return	$(u, c) \mapsto (p, b) \mapsto (q, y)$	each new state is an actual descendant of the token it replaces; each replacement is strict
marked tail $D \geq 3$	$y \mapsto r$	r is a child of the retained LOSS state y, so $h(r) \leq h(y) - 1$
terminal exponent 2/3	$\ell \mapsto Z_2$ or Z_3	the reconstructed terminal token is a proper proof-tree descendant of ℓ
zero-recycle exponent 3	$y \mapsto r$	r is a child of the retained LOSS token y

Table 2: Token replacements used by the strict routing rows. A branching exceptional diamond may replace one token by several descendants; in that case Lemma 4.1 applies to the resulting multiset.

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